

Genome Assembly Research Network

Introduction

A national bioinformatics lab is working on a large genome assembly project. Different teams are responsible for collecting DNA reads, building de Bruijn graphs, detecting motifs, validating assembled genomes, and storing final reports. Because the teams are located in separate lab units, the institute needs a reliable network so that researchers can share data, access web resources, and communicate by email.

You have been appointed as the network engineer of the **Genome Assembly Research Network**. Your task is to design and implement the complete network in Cisco Packet Tracer.

The network consists of six research units:

Unit	Abbreviation	Required Hosts	Description
Sequencing Core Lab	SCL	520	Generates raw DNA reads
Read Mapping Unit	RMU	300	Aligns reads to reference genomes
De Bruijn Graph Unit	DBG	220	Builds assembly graphs
Motif Discovery Unit	MDU	160	Searches regulatory patterns
Genome Validation Unit	GVU	90	Checks assembled genome quality
Archive Storage Node	ASN	45	Stores final genome datasets

Topology Design and Server Setup

1. Each unit must be represented by one router.
2. Each unit must contain:
 - At least two PCs
 - One printer
 - One switch
3. Sequencing Core Lab is the central unit and connects directly to:
 - Read Mapping Unit
 - De Bruijn Graph Unit

- Genome Validation Unit
- 4. Read Mapping Unit, De Bruijn Graph Unit, and Motif Discovery Unit are connected through a shared backbone switch.
- 5. Motif Discovery Unit connects directly to Genome Validation Unit.
- 6. Archive Storage Node connects only to Genome Validation Unit.
- 7. De Bruijn Graph Unit must also have a backup link to Motif Discovery Unit.
- 8. Sequencing Core Lab must host:
 - Central DNS server
 - Web server
 - One email server
 - One DHCP server
- 9. The web server must be reachable from all networks using: www.genome-lab.bio

The homepage must display: Genome Assembly Research Network Online

- 10. Motif Discovery Unit must host another email server.
- 11. Email communication must be configured between two different domains:
mail.sequence.bio and mail.motif.bio
- 12. A user from Sequencing Core Lab must be able to send email to a user from Motif Discovery Unit and vice versa.

Addressing Requirement

The base network address must be derived from the student IDs of the first two members of the group.

Use this rule:

- Take the last two digits of Member 1's student ID. This becomes the first octet.
- Take the last two digits of Member 2's student ID. This becomes the second octet.
- The third and fourth octets are 0.
- The prefix is /16.

So, the base network will be:

X.Y.0.0/16

Example:

Member 1 ID = 20301456 → last two digits = 56

Member 2 ID = 20301729 → last two digits = 29

Base network = 56.29.0.0/16

Apply VLSM subnetting in the correct order. Students must show the full VLSM tree.

DHCP Requirement

Both router-based DHCP and server-based DHCP must be used.

1. Sequencing Core Lab must use a dedicated DHCP server.
2. The Sequencing Core Lab router must use DHCP relay/IP helper for its LAN.
3. Read Mapping Unit and De Bruijn Graph Unit must use router-based DHCP.
4. Motif Discovery Unit must use its own dedicated DHCP server.
5. Genome Validation Unit must receive DHCP service from the Sequencing Core Lab DHCP server using DHCP relay.
6. Archive Storage Node must use static IP addressing only.
7. All routers, servers, and printers must use static IP addresses.
8. Only PCs should receive IP addresses dynamically.

DHCP pool starting addresses:

- Read Mapping Unit DHCP pool must start from the host offset equal to the last two digits of Member 1's student ID.
- De Bruijn Graph Unit DHCP pool must start from the host offset equal to the last two digits of Member 2's student ID.
- Other DHCP pools may start after excluding router, server, and printer addresses.

Routing Requirement

Summary static routes, and fully specified static routes must not be used unless explicitly mentioned below.

The following static routing types must be configured:

1. **Next-hop static route**
Sequencing Core Lab must use a next-hop static route to reach the Archive Storage Node LAN.
2. **Exit-interface static route**
Archive Storage Node must use exit-interface static routing to reach all other networks through Genome Validation Unit.
3. **Recursive static route**
Read Mapping Unit must use a recursive static route to reach the Motif Discovery Unit LAN.
4. **Default static route**
Archive Storage Node must use a default static route toward Genome Validation Unit.
5. **Floating static route**
De Bruijn Graph Unit must have a floating static route to the Sequencing Core Lab LAN through Motif Discovery Unit.
Administrative distance must be 123.

Dynamic Routing Requirement

1. Use RIPv2 for dynamic routing.
2. Dynamic routing must run among:
 - Sequencing Core Lab
 - Read Mapping Unit
 - De Bruijn Graph Unit
 - Motif Discovery Unit
 - Genome Validation Unit
3. Archive Storage Node must not participate in dynamic routing.
4. Genome Validation Unit must make the Archive Storage Node network reachable to the rest of the network.
5. At least one router must use both static and dynamic routing.
6. Sequencing Core Lab must know all networks either dynamically or statically.
7. All devices must be able to ping each other.
8. If the primary route from De Bruijn Graph Unit to Sequencing Core Lab fails, the floating static route must become active.

Deliverables

Students must submit:

1. Cisco Packet Tracer file.
2. Network topology screenshot with labels.
3. Full VLSM tree.
4. IP addressing table.
5. Router configuration commands for all routers.
6. Server configuration screenshots:
 - DNS
 - Web
 - DHCP
 - Email
7. Work distribution table.
8. Short explanation of:
 - Student ID to base network conversion
 - DHCP design
 - Static routing design
 - Dynamic routing design
 - Which router uses both static and dynamic routing

You may make any valid and necessary networking assumptions that are not explicitly stated above, provided they do not conflict with, remove, or simplify any of the listed requirements.